

A PROJECT SYNOPSIS ON

**Real-Time Sign Language Recognition and Translation System Using
Computer Vision**

Submitted in partial fulfilment of the requirements of Project Based Learning (PBL)

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1. Introduction / Background

Communication is a fundamental human need, yet millions of Deaf and hard-of-hearing individuals worldwide face daily barriers because the majority of hearing people do not understand sign language. Sign language is a rich, visually articulated language with its own grammar and syntax, expressed through hand shapes, motion trajectories, and spatial positioning. Despite its expressive completeness, the absence of a common ground between signers and non-signers continues to isolate the Deaf community from mainstream social, educational, and professional interactions.

Advances in computer vision and deep learning have opened new possibilities for bridging this communication gap. Modern hand-tracking frameworks can extract precise skeletal landmarks from ordinary video input, while sequence-based neural networks can learn to recognize the temporal patterns that distinguish one gesture from another. This project explores how these technologies can be combined to build an accessible, real-time sign language translation system that works on everyday hardware, without requiring specialized sensors.

2. Objectives of the Project

- To design a system that captures and interprets hand gestures in real time using a standard webcam feed.
- To extract accurate 3D hand landmark data and normalize it for invariance to distance, camera angle, and hand position.
- To recognize dynamic, continuous sign gestures rather than isolated static poses, by analyzing motion over time.
- To ensure the entire recognition pipeline runs locally on-device, preserving user privacy and eliminating dependency on internet connectivity.
- To build a reusable data-collection pipeline that allows the system's sign vocabulary to be extended by end users.
- To keep the solution affordable and accessible by requiring no specialized hardware such as sensor gloves or depth cameras.

3. Problem Statement

Deaf and hard-of-hearing individuals face severe daily communication barriers because most people do not understand sign language. Existing translation options are often inaccessible: human interpreters are costly and unavailable for spontaneous, everyday interactions, while

current technological solutions typically require bulky sensor gloves or expensive depth cameras. This creates a significant accessibility gap, leaving many Deaf individuals without a convenient, affordable, and immediate means of communicating with the hearing population. There is a clear need for a low-cost, hardware-agnostic system capable of translating sign language gestures into readable output in real time.

4. Scope of the Project

The project will cover:

- Real-time detection and tracking of a single signer's hand(s) using a standard 720p webcam.
- Recognition of dynamic gesture sequences (approximately 1 second / 30 frames long) rather than only static alphabet signs.
- A pipeline to record, label, and export custom gesture data for training new signs.
- An on-device inference engine that requires no cloud connectivity or external API calls.

5. Key Features

- **Real-Time Hand Skeleton Tracking:** Extracts 21 3D hand keypoints per frame from a standard webcam feed, enabling precise, markerless tracking of finger and palm positions.
- **Wrist-Relative Coordinate Normalization:** Zero-centers and scales landmarks relative to the wrist, making recognized gestures invariant to the signer's distance from the camera and to camera angle.
- **Dynamic Motion Recognition:** Processes 30-frame sequence buffers (~1 second of motion) so that fluid, moving gestures are translated correctly, rather than being limited to static letter signs.
- **On-Device & Private Execution:** Runs all computer vision tracking and neural network inference locally, without transmitting video streams over the internet — ensuring user privacy and low latency.
- **Hardware Agnostic:** Operates smoothly on standard 720p webcams with zero specialized hardware or wearable sensors required, keeping the solution low-cost and widely accessible.
- **Dataset Recording Pipeline:** Includes a built-in tool to capture and export 30-frame NumPy array (.npy) sequences, allowing easy training and expansion of a custom sign vocabulary.

6. Methodology / Approach

Phase 1: Data Acquisition

Live video frames are captured from a standard webcam. Each frame is passed through a hand-tracking model that detects 21 3D keypoints (landmarks) corresponding to the joints and tips of the fingers and the wrist.

Phase 2: Preprocessing and Normalization

The extracted landmarks are zero-centered relative to the wrist joint and scaled to a consistent range. This wrist-relative normalization removes variance caused by the signer's distance from the camera, hand size, or camera angle, ensuring that the same gesture produces consistent numerical input regardless of how or where it is performed.

Phase 3: Sequence Buffering

Normalized landmark frames are collected into rolling buffers of 30 consecutive frames (approximately one second of motion at standard frame rates). This allows the system to capture the temporal shape of a gesture rather than a single static pose.

Phase 4: Model Training and Inference

A sequence-based neural network (such as an LSTM/GRU or temporal convolutional model) is trained on labelled 30-frame sequences to classify dynamic gestures. During real-time operation, the trained model performs inference on-device on each new buffered sequence to output the predicted sign.

Phase 5: Dataset Expansion Pipeline

A dedicated recording tool allows new gesture sequences to be captured, labelled, and exported as .npy files, enabling the training vocabulary to be extended with additional signs over time.

Phase 6: Testing and Evaluation

The system will be evaluated on recognition accuracy, latency, and robustness to variations in hand position, lighting, and signer, using a held-out test set of recorded gesture sequences.

7. Expected Outcome / Deliverables

- A functional real-time sign language recognition application that translates hand gestures captured via webcam into readable text output.
- A trained neural network model capable of classifying a defined vocabulary of dynamic sign gestures.

- A reusable dataset-recording tool that allows the sign vocabulary to be extended without modifying the core codebase.
- A fully on-device, privacy-preserving pipeline that requires no specialized hardware, making sign language translation more accessible and affordable.

The broader impact of this project is to lower the barrier to spontaneous communication between Deaf/hard-of-hearing individuals and non-signers, offering a low-cost alternative to human interpreters or expensive sensor-based systems.

8. Literature Review / Existing Systems

Existing sign language recognition approaches broadly fall into two categories. Sensor-based systems use instrumented gloves fitted with flex sensors and accelerometers to capture hand movement directly; while accurate, they are expensive, intrusive, and impractical for everyday casual use. Vision-based systems using depth cameras (e.g., Kinect-style sensors) improve accessibility but still require dedicated hardware beyond what most users own. More recent vision-based approaches using standard RGB webcams and 2D/3D hand-landmark estimation have shown promise but often focus on static alphabet recognition rather than fluid, continuous gestures, limiting their practical usefulness.

This project differentiates itself by combining wrist-relative landmark normalization with temporal sequence modelling, enabling recognition of dynamic, continuous signs using only a standard webcam — removing both the hardware cost barrier and the static-gesture limitation seen in prior accessible solutions.

9. References / Bibliography

Research papers, articles, and documentation referred to during the project will be listed here in standard citation format (e.g., IEEE), such as:

- Official documentation of the hand-tracking framework (e.g., MediaPipe Hands) used for landmark extraction.
- Peer-reviewed papers on sign language recognition using deep learning and skeleton-based approaches.
- Documentation for the deep learning framework used (PyTorch).

- Relevant datasets or benchmark studies on gesture and sign language recognition, if used for reference.